

LBIST Project — File-by-File Explanation

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This document describes every file in the LBIST project directory, grouped by purpose. Each entry lists what the file is and its role in the design flow.

1 RTL Source Files

pattern_gen.sv *Test Pattern Generator*

A 4-bit maximal-length Linear Feedback Shift Register (LFSR) with taps `out[3] ⊕ out[0]` and seed 0001. Produces the 15 test vectors that exercise the CUT, achieving full period $2^4 - 1 = 15$.

cut.sv *Circuit Under Test*

Combinational CUT: `out = (in[0] & in[1]) | (in[2] & in[3])`. Synthesizes to 3 NAND gates generically, or a single SKY130 `a22o_1` cell.

stuck_models.sv *Fault-Injection Model*

Combinational model producing the golden output and 12 stuck-at-fault variants. The active fault is selected by the 4-bit `smodel` input.

MISR.sv *Multi-Input Signature Register (ORA)*

Contains 13 parallel 4-bit MISRs (12 faulty + 1 golden). Each compresses a 15-bit response stream into a 4-bit signature. Total: 52 flip-flops + 26 XOR gates.

comparator.sv *Signature Comparator*

Compares each faulty signature against the golden signature. Emits 12 detection flags and a 4-bit `num_detected` count.

controller.sv *BIST Controller FSM*

4-state FSM (IDLE → RUN → EVAL → DONE). Controls the LFSR enable, snapshot timing, and completion. Emits `done`, `pass`, `coverage`.

scan_ff.sv *Scan Flip-Flop*

DFT-ready flip-flop with shift, capture, and hold modes. Provides TDI/TDO chain access.

scan_reg.sv *Scan Chain*

Chain of `REG_SIZE` scan flip-flops. Part of the DFT infrastructure; not yet wired into the BIST top.

d_ff.sv *D Flip-Flop Primitive*

Basic D flip-flop with async active-low reset. Building block for the scan flip-flop.

top.sv *BIST Top-Level*

Integrates LFSR, CUT, fault model, MISR, comparator, and controller into a single synthesizable hierarchy.

tb.sv *Testbench*

Self-checking testbench that sweeps all 12 fault models, reports coverage, and dumps VCD.

2 Generic Synthesis and Netlist Files

cut_netlist.v *Generic CUT Netlist*

Gate-level netlist of `cut.sv` produced by Yosys. Contains 3 `$_NAND_` primitives. Used as input to AU-COHL/Fault.

netlist.v *Generic Full BIST Netlist*

Complete gate-level netlist of the BIST top. ~380 generic cells.

simcells.v *Yosys Cell Simulation Library*

Verilog models for Yosys primitives (`$_AND_`, `$_NAND_`, `$_DFF_PNO_`, etc.). Passed to AU-COHL/Fault as the cell model.

simlib.v*Yosys Full Simulation Library*

Larger library for complex cells (`$add`, `$mux`, `$dff`, `$mem`). Used when the design has higher-level primitives.

3 SkyWater SKY130 PDK Synthesis Files

cut_netlist_sky130.v*SKY130 CUT Netlist*

Gate-level netlist of `cut.sv` mapped to **1 SKY130 a22o_1** cell, with area 8.76 μm^2 . Demonstrates technology-aware compound-cell mapping.

bist_netlist_sky130.v*SKY130 Full BIST Netlist*

Complete BIST design mapped to the `sky130_fd_sc_hd` high-density standard cell library. Contains **348 standard cells** across 21 distinct cell types. Total area: **3905 μm^2** .

sky130_fd_sc_hd_tt_025C_1v80.lib*SKY130 Liberty File*

Liberty timing file describing every cell in the SKY130 high-density library. Process corner: typical-typical, 25°C, 1.8V. Used by Yosys for `dfflibmap` and `abc` technology mapping.

synthesis_report_sky130.txt*SKY130 Synthesis Log*

Full `stat -liberty` output showing per-module cell counts and area breakdowns for the entire SKY130 synthesis run.

sky130_cells.v*Minimal SKY130 Cell Model for Fault Simulation*

A hand-written Verilog model defining the `sky130_fd_sc_hd__a22o_1` compound cell for AUCOHL/Fault. Required because the standard `simcells.v` library only defines generic Yosys primitives, not real SKY130 cells.

faults_sky130.yml*SKY130 Fault List*

The 10 fault sites enumerated by Fault in the SKY130-mapped CUT, including the A22O cell's 4 inputs and output, plus the 5 primary I/O ports.

covered_sky130.yml*SKY130 Baseline Coverage Report*

Coverage achieved with Fault's internal random generator: **100%**.

lfsr_covered_sky130.yml*SKY130 LBIST Coverage Report*

Coverage achieved when replaying the LBIST's 15 LFSR patterns on the SKY130 netlist: **100%**. Confirms the LBIST is effective on real PDK-mapped netlists.

4 Fault Simulation Artifacts (AUCOHL/Fault)

cut_netlist.v.tv.json*Auto-Generated Test Vectors*

Industrial simulator's internally generated test patterns in JSON, with per-vector coverage metadata.

cut_netlist.v.tv.svf*Serial Vector Format Output*

Test vectors exported in SVF (ATE-compatible format).

faults.yml*Exhaustive Fault List*

14 fault sites (28 stuck-at faults) enumerated by Fault in the synthesized CUT.

covered.yml*Baseline Coverage Report*

Coverage achieved by Fault's internal random generator: **100%**.

lfsr_patterns.json*LFSR Patterns Encoded for Fault*

15 LFSR patterns re-encoded into Fault's JSON format. Fed via `-externalTVSet`.

lfsr_covered.yml*LBIST Coverage Report*

Coverage achieved by replaying our LFSR's 15 patterns: **100%**.

parsetab.py*Auto-Generated Parser Table*

Python parser table auto-generated by AUCOHL/Fault's Yacc-based parser.

5 Simulation and Waveform Files

bist.vcd	<i>Waveform Dump</i>
VCD capturing all signal transitions during simulation. Contains the full FSM trace, LFSR patterns, MISR signatures, and detection flags.	
build.log	<i>Compilation Log</i>
Output from Icarus Verilog compile.	
tb.vvp	<i>Compiled Testbench Binary</i>
Compiled simulation binary for the testbench.	
top.vvp	<i>Compiled BIST Binary</i>
Compiled simulation binary for the top-level BIST alone.	

6 Summary

Category	Count	Purpose
RTL source files	10	Original SystemVerilog design
Generic synthesis / netlists	4	Gate-level outputs + cell libraries
SKY130 PDK files	4	Liberty file, PDK netlists, synthesis log
Fault-sim artifacts	7	Coverage reports, patterns, fault lists
Simulation / waveform	4	Compiled binaries, VCD, build log

Total: 29 files covering the complete LBIST design, generic synthesis, SKY130 PDK synthesis, functional verification, and fault-coverage evaluation flow.